A Study on Exploring and Exploiting the High-dimensional Design Space for Analog Circuit Design Automation

(Invited Paper)

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2024 29th Asia and South Pacific Design Automation Conference (ASP-DAC) | 979-8-3503-9354-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/ASP-DAC58780.2024.10473920

Abstract— The escalated intricacy of analog circuits, compounded by the high-dimensional nature of the design space, introduces complexities in optimizing circuit performance. Since the evaluation cost, often through circuit simulation, is resource-intensive and time-consuming, it is crucial to obtain a feasible design with a decent Figure of Merit (FOM) value within a limited simulation budget. In this study, we conduct an in-depth review and analysis of cutting-edge exploration and exploitation techniques developed to address the intricacies encountered in analog circuit design automation. Moreover, to enable algorithmic comparisons and advance the state of the field, we provide benchmarks encompassing analog circuit netlists with high-dimensional design variables, which empower researchers to rigorously assess and refine their optimization algorithms, leading to enhanced efficacy and novel developments.

I. Introduction

Electronic Design Automation (EDA) is the cornerstone of modern integrated circuit development. After the 1950s, with the maturation of device physics theory and circuit analysis theory, EDA and integrated circuit design entered a rapid development phase [1]. Today, however, for analog integrated circuits, their design still relies on human expertise, which is the major bottleneck in the time-to-market cycle [2]. Hence, achieving efficient and agile design for analog integrated circuits is currently an urgent and crucial problem to be addressed.

The automation of analog integrated circuit design encompasses tasks such as circuit topology optimization, circuit design parameter optimization, and automated circuit layout. This study focuses on the problem of design parameter optimization driven by circuit performance. In other words, it involves automatically determining feasible design parameters for a given circuit topology and performance specifications. These design parameters may include transistor dimensions, finger count, passive component values, bias values, and so on. The design parameter optimization problem for circuits can be mathematically abstracted as a high-dimensional constrained nonlinear optimization problem.

From a modeling and optimization standpoint, analog integrated circuit optimization methods fall into two categories: **knowledge model-based** and **black-box model-based**. Knowledge-based methods rely on designers' expertise, while black-box methods treat the problem as an optimization challenge without specific knowledge. These approaches are suitable for design variables with fewer than 50 dimensions. However, when tackling high-dimensional challenges in growing analog circuits, their shortcomings become more pronounced, while their benefits diminish. For example, polynomial model-based algorithms are fast but have low predictive accuracy. Genetic evolution-based algorithms yield high-quality points but suffer from an overwhelming number of simulations. Bayesian optimization with Gaussian processes brings exponential computational complexity. The critical concern is these methods prioritize finding the optimal design point but may disregard iteration costs. High-dimensional analog circuit simulations are time-consuming, necessitating efficient solutions in limited iterations [3]. In essence, effective algorithms for high-dimensional analog circuit optimization must strike an exploration-exploitation balance.

In recent years, to address design optimization problems with a large number of free design variables, new high-dimensional optimization methods based on dimension reduction and region planning have been proposed. This mainly includes dimension decoupling, dimension embedding, region restriction, and region decomposition.

Challenges persist in applying these methods to analog circuit optimization. The circuit objective function is frequently highly singular, involving multiple competing objectives like gain and bandwidth. Some efforts combine region planning techniques to identify feasible solution areas. Within these regions, circuit responses are often sensitive to just a few parameters, making dimension reduction beneficial, reducing complexity, and expediting optimal solutions. However, region restriction adds computational overhead, and subspaces within feasible regions grapple with dimensionality issues. More efficient exploration-exploitation algorithms warrant exploration.

Another significant constraint is the absence of a unified metric. To address this, we have designed an open-source benchmark: **ACOB**, using analog circuits provided by the IEEE Standard Association. ACOB covers circuit optimization problems ranging from low-dimensional to high-dimensional, enabling researchers to evaluate their algorithms. We will continue to maintain and expand this benchmark with more state-of-the-art analog circuits in the future.

The rest of this article is organized as follows. Section II

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introduces the analog device sizing problem and the challenges faced. Section III introduces the mainstream methods for low dimensional circuit optimization. Section IV introduces the state-of-the-art algorithm for the high dimensional optimization problem, and Section V discusses the application of these methods in analog circuit optimization. Section VI briefly introduces our open-source benchmarks, and SectionVII summarizes the paper.

II. Problems and High-dimensional Challenges

A. Analog Device Sizing Problem

An analog circuit sizing problem can be formulated as a constrained optimization problem,

$$\begin{aligned} &\underset{x \in \boldsymbol{\chi}^{\mathcal{D}}}{\text{minimize}}\ \boldsymbol{f}_m(\boldsymbol{x}) \\ &\text{s.t. } c_i(x) < 0 \\ &\forall i \in 1 \ldots N_c, \end{aligned} \tag{1}$$

where $\boldsymbol{x}$ denotes design parameters, such as W, L of transistors. $\chi^{\mathcal{D}}$ denotes the design space composed of upper and lower bounds of $\mathcal{D}$ variables. $\boldsymbol{f}_m(\cdot)$ denotes the objective function vector. If $m = 1$, it's a single objective optimization problem, and the objective function is usually the Figure of Merit (FOM). A FOM is the combination of the analog circuits' performances with which the designer is concerned. c_i denotes the ith constrain, and N_c denotes the number of constrains. Also, there can be multiple objective functions. For multi-objective optimization, the trade-off of various performances should be considered. Because it's hard to evaluate two points, each with advantages of different performances. The goal is to find a set of solutions to approximate the $ParetoFront(PF)$. A solution x_i is on the PF means that x_i can not be $dominated$ by any other solutions [4]. x_1 dominates x_2 as follows:

$$\begin{aligned} &\forall i\ f_i(x_1) \leq f_i(x_2), i \in \{1, ..., m\} \\ &\exists j\ f_j(x_1) < f_j(x_2), j \in \{1, ..., m\} \end{aligned} \tag{2}$$

B. High-dimensional Challenges

The automated design of analog integrated circuits faces significant challenges, including strong nonlinearity in circuit performance functions, high dimensionality of free design variables, and long simulation times. Existing methods exhibit exponential complexity with respect to dimensionality, making it difficult to address high-dimensional problems with limited resources.

Strong nonlinearity: The response functions of circuit performance are often discontinuous. Taking the example of an oscillator circuit, it does not oscillate over the majority of the design parameter space. Therefore, optimization algorithms for analog integrated circuit automation need to be capable of automatically exploring the feasible design space in high dimensions. To gain an intuitive understanding, we take the width of two transistors in an OP-AMP circuit as free variables. We use the FOM composed of power, delay, offset, and output voltage peak-to-peak, weighted and summed, as the objective function [5]. We randomly sample 900 times, resulting in the design space as shown in Fig.1.

Massive free variables: With the development of Moore's Law, the scale of analog integrated circuit systems has grown

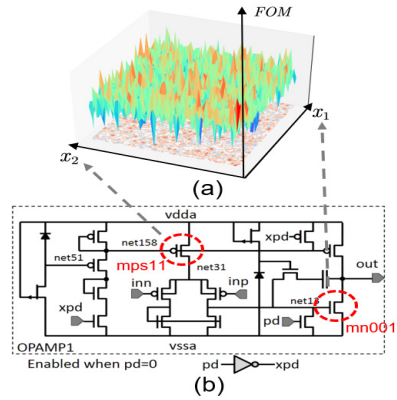

Fig. 1. Design space of an OP-AMP circuit. We select the width of $mn001, mps11$ as independent variables, corresponding to x_1, x_2. FOM is the objective function, which consists of a weighted sum of power, delay, offset, and output voltage peak-to-peak.

from a few dozen transistors in the early days to several thousand or even tens of thousands of transistors today. The number of free design variables available for tuning has grown exponentially. The growth results in an exponential increase in the design space. Therefore, optimization algorithms for analog integrated circuit automation need to be capable of handling high-dimensional optimization problems.

Time-consuming simulation: Taking commonly used circuits such as Analog-to-Digital Converters (ADCs) and Phase-Locked Loops (PLLs) as examples, a single simulation for noise, multi-process corners, and other design metrics validation can take several hours. Since the design automation relies on simulation, the extended duration of a single simulation greatly restricts the number of optimization iterations within a given time constraint. Therefore, optimization algorithms for analog integrated circuit automation need to obtain feasible circuit designs with a minimal number of iterations. In summary, the core scientific problem in the automated optimization of analog integrated circuits is how to automatically match the feasible design space in high-dimensional design space and obtain design parameters that meet the requirements of circuit performance with a limited number of simulations.

In general, when addressing the high-dimensional challenges of analog circuit optimization, the algorithms should have improved global exploration strategies to prevent falling into local optima. They must also consider the cost of iterations rather than merely pursuing the global optimum. Therefore, the key lies in balancing the **exploration-exploitation** trade-off.

III. When Mainstream methods meet High-dimensional Challenges

In this section, we briefly reviewed classical analog circuit optimization algorithms, analyzed the characteristics of each algorithm, and identified their limitations in addressing high-dimensional optimization challenges. Table I. summarizes the features of these algorithms.

A. Knowledge-based Methods

Methods based on Knowledge models [6]–[17] utilize either human design experience or the derivation of circuit transfer

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equations to establish mathematical models with analytical forms that describe the relationship between design parameters and circuit performance in analog integrated circuits. Alternatively, these methods involve regression modeling of observed simulation data points, with optimization techniques used to find the optimal design parameters. Such methods involve a one-time offline modeling of the circuit and do not update the model during the optimization process.

Hachtel et al. [16] were early adopters of the gradient descent method for sizing optimization. They initially used a direct method to compute gradients, which allowed obtaining one parameter's gradient after each simulation. Subsequently, [17] introduced an adjoint method, saving all parameter gradients after a single simulation. However, time-domain optimization added convolution, increasing computational complexity. JiffyTune enhanced gradient efficiency with an adjoint Lagrangian method [14]. Despite these improvements, gradient-based methods still grapple with challenges, including high computational complexity and sensitivity to initial conditions, which can lead to local optima convergence.

The Geometric Programming method, introduced in [6] [7], formulates analog integrated circuit optimization as convex problems. It creates mathematical expressions for design parameters, intermediate variables, and circuit performance, solved using numerical optimization. Leveraging the favorable characteristics of convex optimization, like differentiability and global convergence, this method efficiently finds the global optimum. However, mathematical modeling relying on human expertise may struggle to describe complex circuit behaviors such as parasitic effects and multi-process corner deviations. The modeling predictions often exhibit significant discrepancies with SPICE simulations.

In summary, knowledge-based methods rely on expert insights to create a precise model linking circuit performance and design parameters. Once a dependable model exists, these methods offer high optimization efficiency and ensure global optimal designs. However, the variability in analog integrated circuit topologies and complex behavior makes establishing accurate relationships challenging. Manual derivation or experimental fitting can lead to significant disparities between optimization results and actual circuit behavior, restricting their suitability to small-scale circuits.

B. Black Box-based Methods

Black Box-based methods [18]–[29] drive circuit optimization directly through circuit parameters. They evaluate the circuit's performance for candidate points through simulations and use optimization strategies to find the global optimum. Depending on their strategies, black-box methods can be categorized as 'offline-based' and 'online-based'. Online methods continuously update their surrogate model during optimization, while offline methods do not.

1) Offline-based Methods: These methods directly invoke optimization algorithms to iteratively generate circuit candidate points and do not establish surrogate models for circuit behavior.

A method based on positive polynomial models using simulated analog integrated circuit performance was introduced in [19]. This method fitted the parameters of the given positive polynomial model with simulated sample data, improving modeling accuracy. [18] proposed an optimization method based on sparse polynomial models and semi-definite programming. By applying sparse constraints to polynomial coefficients, the modeling cost was reduced. However, fitting polynomial parameters using circuit simulation data often requires a large number of simulation points, resulting in high modeling costs.

Georges Gielen et al. were among the first to apply simulated annealing to address analog integrated circuit optimization problems [30]. This method adjusts the 'annealing temperature' parameter to control optimization algorithm perturbations and escapes local optima. However, for multi-modal analog integrated circuit optimization, simulated annealing exhibited numerical instability. A higher 'annealing temperature' increased circuit simulations, challenging convergence, while a lower temperature risked local optima, yielding suboptimal results.

Genetic Programming is used in the automated optimization of analog integrated circuits [20]. It generates circuit candidate points through population evolution mechanisms. However, the determination of the optimal point depends on a certain 'population' size and evolution generation, which makes genetic algorithms relatively slow in achieving global convergence.

Liu et al. proposed an improved Differential Evolution(DE) method (Competitive Co-Evolutionary Differential Evolution) for optimizing analog integrated circuits [21]. Compared to genetic algorithms, this method achieved improved search efficiency and global convergence, but it required a large number of simulations, resulting in significant time overhead.

The Particle Swarm Optimization(PSO) method, introduced by Vural et al. [22], employs agent modeling and interaction variables to enhance diversity in direction selection during optimization, aiming for improved exploration of the global design space optimum. With fewer mathematical operators than genetic algorithms, such as mutation and crossover, it offers computational efficiency. However, its search performance falls short, providing only approximate global optimum solutions.

Qian et al. proposed the Multiple Starting-Point(MSP) approach [23]. This method generated multiple starting points in the design space, initializing local optimization algorithms based on gradients. Compared to the aforementioned methods, it improved the global convergence and convergence speed of the optimization algorithm.

In summary, simulation-based analog integrated circuit optimization methods do not model circuit behavior but employ iterative circuit simulations to search for optima. Common strategies include gradient-based and evolutionary algorithms, which provide accurate results by directly invoking SPICE simulations. However, high-dimensional circuits entail time-consuming single-time simulations. Conducting a large number of simulations for optimization can significantly prolong the process, often extending it to several months.

2) Online-based Methods: Online-based optimization methods update surrogate models throughout the optimization process. These models guide optimization algorithms in selecting candidate circuits (see Fig.2 for a simple online

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model). The purple curve represents the objective function, typically not directly obtainable, while the solid blue line is the surrogate model based on known observations. By optimizing the surrogate model using the acquisition function, the next exploration point is determined. The blue-shaded area indicates the confidence interval; regions with more exploration have a smaller confidence interval, bringing the surrogate model closer to the target function. These methods significantly reduce the number of circuit simulation points during optimization, resulting in time savings. In recent years, online model-based optimization methods using machine learning [31]–[33], deep learning [34]–[36], and Bayesian optimization [37]–[43] has gradually become a hot topic in analog integrated circuit optimization.

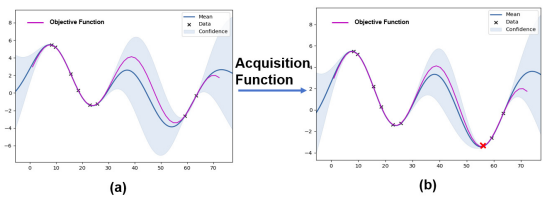

Fig. 2. An example of a Bayesian online model: The purple curve signifies the objective function (typically not directly accessible), while the solid blue line represents the surrogate model trained from available data. Optimization of the surrogate model through the acquisition function identifies the next exploration point, denoted by a red cross in **(b)**. The blue shaded area represents the confidence interval. Notably, regions with increased exploration yield smaller confidence intervals, indicating a closer approximation of the surrogate model to the target function.

Machine learning methods often create surrogate models using artificial neural networks. Li et al. introduced a method based on Artificial Neural Network (ANN) models [31]. These models were trained using circuit simulation data for optimization. Budak et al. introduced a neural network-assisted Differential Evolution algorithm [32]. Their surrogate models improved prediction accuracy and reduced sample size requirements by incorporating data augmentation techniques, addressing traditional neural network limitations.

Reinforcement learning methods iteratively introduce sample points to update the 'action-value network.' In this context, the 'action network' generates candidate design parameters, and the 'value network' evaluates and corrects the 'action' model's predictions. AutoCkt introduced Sparse Sampling Reinforcement Learning [34], using discrete actions in the design space for circuit parameter optimization, achieving better efficiency than random reinforcement learning and DE. However, it still required thousands of SPICE simulations for training before model deployment, incurring substantial simulation costs. Budak et al. introduced DNN-Opt [35], which improved optimization efficiency compared to DE algorithms.

Bayesian optimization(BO) methods use Gaussian process(GP) models as online models, regressing the circuit performance function's distribution parameters with simulation data. The online model is optimized to find new iteration points until convergence to the optimal design point is achieved. Liu et al. introduced the GASPAD method [37], which improved DE in millimeter-wave integrated circuit synthesis using GP models. GASPAD enhanced diversity and effectiveness in global optimization by incorporating the best offspring population from DE into sample points for GP modeling. This significantly reduced simulation points while maintaining optimization quality. Zhang et al. [39] proposed BO with deep neural network models, reducing modeling and prediction complexity from $O(n^3)$ to $O(n)$. Sparse Gaussian modeling approximated traditional GP models using inducing points, enhancing generality. Parallel BO methods [41]–[43] introduced various batch selection mechanisms based on multi-objective optimization, extending BO from serial to parallel optimization.

The aforementioned research results demonstrate that BO methods are highly effective for small-scale analog integrated circuit optimization problems. However, as circuit dimensions increase, GP model accuracy decreases, and exploration efficiency is limited.

In summary, online model-based analog circuit optimization employs machine learning, deep learning, and Bayesian techniques to create surrogate models, resulting in reduced simulation points and optimization time. Yet, in high-dimensional design spaces, these methods face limitations, including sparse sample points, complex model training, and reduced accuracy with increased dimensions. As a response, recent approaches have emerged, focusing on high-dimensional analog circuit optimization through dimension reduction and region planning.

IV. Cutting-edge High-dimensional Optimization Approaches

With the continuous expansion of analog integrated circuit complexity, the search space for optimization algorithms grows exponentially with the increase in the dimensionality of design variables, resulting in an exponential increase in problem-solving complexity. To mitigate blind exploration in high-dimensional spaces and improve search efficiency, various dimension reduction and region planning-based optimization methods have been proposed in recent years. These methods primarily include dimension decoupling, dimension embedding, region restriction, and region decomposition.

A. Dimension Decoupling

Methods based on dimension decoupling assume the existence of inherent structural relationships between the dimensions of the optimization problem, such as additivity. Hyperparameters of the dimension structure model are trained using machine learning techniques. In the optimization process, dimension synthesis methods first decouple the high-dimensional design space into multiple low-dimensional spaces, where they model the optimization problem separately in each low-dimensional space. Subsequently, several low-dimensional models are structurally reconstructed and mapped back to the high-dimensional space to find the optimal solution. [44]–[46] proposed the use of Additive Gaussian Process(ADD-GP) models to perform dimension decomposition on high-dimensional optimization problems. They maximized the marginal likelihood of this GP model to obtain model parameters. [47] introduced a new acquisition function called Max-value Entropy Search (MES) on the basis of ADD-GP models. MES maximizes the mutual information gain between the current best value and the next observation point,

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TABLE I
A SUMMARY OF CLASSICAL METHODS FOR ANALOG CIRCUIT OPTIMIZATION

Model Type	Article	Method	Multi/Single Objective	With constraints?	Support Parallelization?	Global/Local	Variables[1]	Others
Knowledge-based	[16]	Gradient	Single	✗	✗	Local	3	Among the first.
	[17]	Gradient+Adjoint sensitivity	Single	✗	✗	Local	6	Suitable for optimizing circuit structure.
	[14]	Adjoint Lagrangian method	Single	✗	✗	Local	60	Designer-friendly interfaces.
	[6]	Geometric Programming	Single	✔	✗	Global	14	None
	[13]	Nonconvex polynomial optimization	Single	✔	✗	Global	19	None
	[18]	Sparse polynomial (non-convex) optimization	Single	✔	✗	Global	11	Leverage progress on Semidenite Programming.
Offline Model	[19]	Posynomial Modeling from simulation points	Single	✔	✗	Global	13	None
	[25]	Evoliutionary Algorithm(EA)	Multi	✔	✔	Global	21	Latin supercube sampling+MOEA/D, achieve 1% accurary with 5000 samples.
	[27]	Particle Swarm Optimization(PSO)	Single	✗	✗	Global	6	None
	[21]	Competitive Co-evolutionary Differential Evolution(CODE)	Single	✔	✗	Global	50	None
	[26]	Stochastic pattern search	Multi	✔	✔	Global	71	Simulate 50000-100000 samples, cost hours.
	[24]	Multiple start point	Single	✔	✗	Global	36	Global exploration+local expolition.
Online Model	[31]	Genetic Algorithm + Artificial Neural Network	Multi	✔	✔	Global	12	For a CBPF circuit, 632 observations are need to achieve 1% accuracy.
	[32]	Differential Evolution + Artificial Neural Network	Single	✔	✗	Global	25	None
	[34]	Reinforce Learning + Transfer learning	Multi	✔	✔	Global	11	10^4 samples for a two stage op-amp.
	[35]	Reinforce Learning + Deep Neural Network	Single	✔	✗	Global	20	Adopt sensitivity analysis.
	[37]	Differential Evolution + Bayesian Optimization(BO)	Single	✔	✗	Global	33	None
	[38]	Bayesian Optimization	Multi	✔	✗	Global	36	None
	[41]	BO+multiobjective-acquisition function	Single	✔	✗	Global	36	None
	[42]	BO+improved acquisition function	Multi	✔	✔	Global	36	None

[1] We conducted the number of variables in the circuit with the most variables in each article's experimental section.

which is used to select the next simulation point. However, these methods rely on strong assumptions about dimension additivity, which may not necessarily hold in practical analog circuit. In particular, the performance of transistors can vary significantly in different operational regions, and the relationships between design parameters are often strongly related to the circuit's operational state. Simple additivity may not be sufficient to describe the complex behavior of analog integrated circuit systems. Moreover, dimension decoupling methods introduce multiple low-dimensional surrogate models, which inevitably increase training complexity and computational costs. Although [48]–[50] proposed various methods to approximate the kernel of GP with Fourier features, extending them to high-dimensional problems is still challenging

B. Dimension Embedding

Methods based on dimension embedding assume that the underlying dimension of the optimization problem is lower than the number of design variables. They use dimension reduction algorithms to extract effective low-dimensional subspaces to avoid the high complexity associated with directly optimizing the original high-dimensional problem. [51], [52] introduced the Random Embedding Bayesian Optimization (REMBO) method, which employed random dimension embedding to project the high-dimensional space onto a lower-dimensional effective subspace. REMBO optimized acquisition functions in this effective subspace, efficiently selecting new candidate points. A Hashing-enhanced Subspace Bayesian Optimization(HesBO) method was proposed by [53], avoiding the need for complicated projection correction operations, thus lowering the computational cost. However, [51]–[54] directly reduce dimension over the entire design space with linear embedding. Since each design parameter in the full space may influence circuit performance to some extent, the dimension reduction effect on circuit problems is limited, leading to subpar optimization performance. Besides, [55], [56] focus on finding the sensitive variables to improve sampling efficiency. Neural Networks are also introduced by [57], [58] to embed the original variables to a non-linear feature space.

C. Region Restriction

Methods based on region constraints utilize means such as trust region radii to progressively narrow the search space during optimization iterations, restricting the optimization algorithm to explore only specific regions within the high-dimensional design space. The Trust Region Bayesian Optimization (TuRBO) method was introduced in [59], which focused the optimization process within a rectangular trust region centered around the current best-observed value. Sobol sampling was used to generate candidate simulation points within the trust region, and the radius of the trust region was dynamically adjusted based on the quality of observed points.

D. Region Decomposition

Methods based on region decomposition dynamically divide the design space into 'good' and 'bad' regions based on the simulations and location information of observed points, thereby restricting the optimization algorithm to operate within specific regions. [60] proposed a Monte Carlo Tree Search (MCTS) method based on clustering observed points and using a support vector machine (SVM) to explicitly define the region boundaries between 'good' and 'bad' regions. The upper confidence bound function was used to balance exploration and exploitation, selecting appropriate target regions for optimization. However, the training complexity of SVM is $O(N_s^3)$, and the computational cost of explicitly defining boundaries increases cubically with the number of observed points, resulting in high costs as the number of subspaces for division increases. [61] introduced the Thresholded Domain Shrinking algorithm, based on a binary tree structure that continually divides the feasible target regions into smaller regions in each dimension. This approach improved search efficiency while reducing computational complexity. However, with each division operation, the depth of the binary tree increased by d (where d is the dimension of the design space), and the number of subdivided subspaces increased

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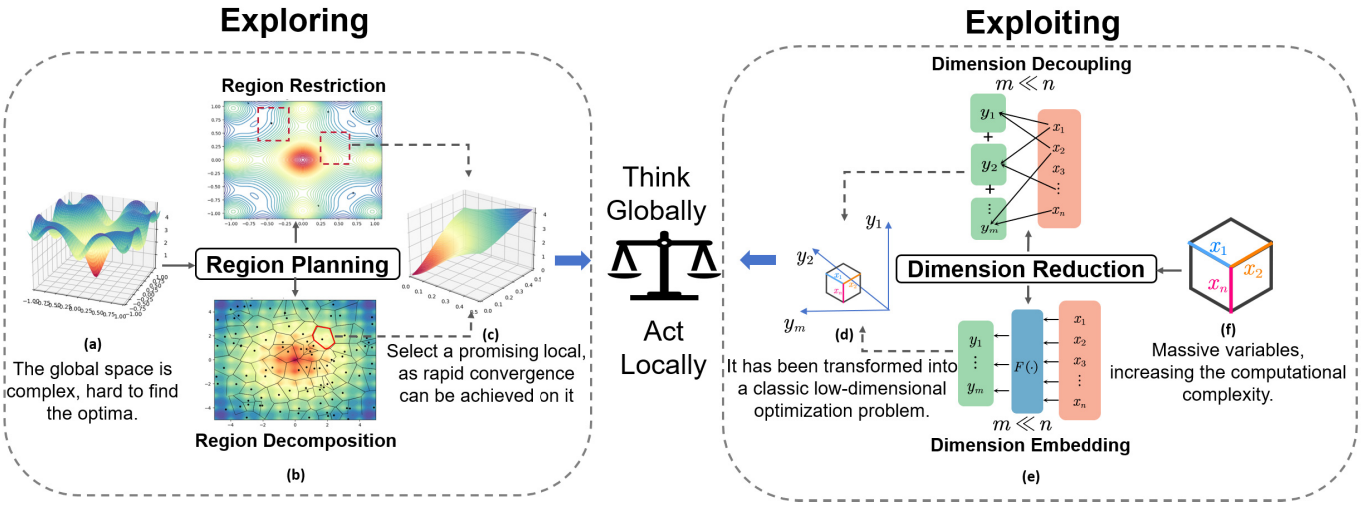

Fig. 3. Conclusion on High-Dimensional Optimization Approaches. On the left, two region planning approaches are introduced: region restriction and region decomposition, as depicted in **(b)**. Region restriction involves selecting promising areas within the entire design space, while region decomposition divides the design space into parts and selects promising ones through methods like tree search. Region planning methods can transform complex design spaces, as shown in **(a)**, into smoother subspaces, as in **(c)**, ensuring algorithm convergence. On the right, two dimension reduction methods are discussed: dimension decoupling and dimension embedding, as seen in **(e)**. Dimension decoupling breaks the high-dimensional variable space into lower-dimensional subspaces, assuming additivity between them. Dimension embedding maps high-dimensional variables to a lower-dimensional feature space using methods like neural networks. After dimension reduction, the space originally spanned by n variables is mapped to a space with m variables, where $m \ll n$, as depicted in **(d)**.

exponentially. Therefore, this method is mainly applicable to low-dimensional problems.

In summary, international approaches for solving high-dimensional optimization problems revolve around dimension reduction and region planning, as shown in Fig. 3. Most of these methods originate from artificial intelligence but lack focus on the nuances of analog integrated circuit optimization. Dimension reduction methods capitalize on inherent dimension relationships, guiding optimization in lower-dimensional subspaces, and substantially reducing computational complexity. However, they rely on prior knowledge or extensive data support, offering no guarantee of consistency with the original problem's optimal solution. Region planning restricts optimization to feasible or potentially global optimal regions within the design space. Yet, it introduces computational overhead, and optimizing subspaces within feasible regions still grapples with the curse of dimension. Building on the above methods, it is possible to use region planning methods to obtain feasible solution regions. Since the response functions of circuit problems within feasible solution regions often exhibit sensitivity to a small number of parameters, dimension reduction within feasible solution regions can yield better results, subsequently reducing optimization complexity and achieving rapid optimal solutions.

V. Applying High-Dimensional Approaches to Analog Circuit Optimization

The core challenge in the automated optimization of analog integrated circuits remains how to automatically match feasible circuit design spaces within the high-dimensional design space and obtain a better design parameters through a minimal number of simulations. To address this issue more effectively, it is necessary to apply state-of-the-art high-dimensional optimization methods in a more targeted manner. Some prior work has attempted to apply the above methods to the problem. These efforts mostly are exploratory, we briefly introduce them below.

Dimension Decoupling. Kandasamy assumes the objective function has an additive structure, i.e. it is a sum of lower dimensional components [62]. This allows the statistical and computational complexity to scale better with dimensionality compared to standard GP-UCB. An additive Gaussian process model is used to handle the high dimensionality from extra LDE variables [3]. Objectives are optimized separately. To reduce the number of samples, the Mondrian process is adopted [63].

Dimension Embedding. Chen et al. adopt a dropout strategy based on mutual information analysis to optimize a subset of selected variables at each iteration [64]. It also introduces a g_m/I_D methodology-based optimization strategy, where optimization variables are g_m/I_D, I_D, and length instead of transistor width and length. BBGP-sDFO integrates a batch Bayesian querying strategy for exploring the global design space and a Gaussian process enhanced subspace derivative-free optimization method for exploiting promising regions in an effective low-dimensional subspace [63]. The subspace is spanned by the approximate gradients and the previous iteration path.

Region Restriction. LoCoMOBO [65] and [66] applied the TuRBO method to the optimization of analog integrated circuit dimensions and achieved some acceleration for circuits with over 20 dimensions compared to traditional Bayesian optimization. However, methods based on region restriction often rely on stochastic mechanisms to initialize trust region positions and can perform poorly for complex multi-modal optimization problems, easily getting stuck in local optima. Allocating more computational resources and performing parallel optimization within multiple trust regions can to some extent escape local optima.

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Region Decomposition. Decomposing the design space and optimizing locally can make the obtained solutions dependent on the choice of the initial region. Zhao et al. [2] use a hierarchical Voronoi tree to progressively decompose the design space into partitions with implicit performance boundaries. This allows promising regions to be explored efficiently. The Voronoi tree is constructed dynamically using a feature-weighted K-Means clustering method to split nodes into 'good' and 'bad' subspaces. The tree is traversed using UCT to select promising leaf nodes.

In summary, based on our classification, methods mentioned above employ only a single strategy for addressing high-dimensional analog circuit optimization problems. In the future, a fusion strategy that combines regional planning and dimension reduction may achieve a balance between exploration and exploitation. Furthermore, it is worth exploring how to leverage circuit theory and artificial design experience to guide optimization algorithms for rapid convergence.

VI. ACOB: Analog Circuit Optimization Benchmarks

We have released an open-source benchmark, ACOB, on GitHub[1], currently featuring three analog circuits: OSC, OPAMP, and PLL. These circuits are sourced from [5]. We have configured 14, 36, and 77 independent variables based on manual design experience for each of them, along with user-friendly simulation calls and parameter modification interfaces. In the future, we will continue to maintain this project and add large-scale system-level circuits to make it more in line with real industrial scenarios.

VII. Summary and Conclusions

Our study provides an overview of mainstream analog circuit optimization approaches and their high-dimensional challenges in the face of growing circuit complexity. We emphasize the exploration-exploitation trade-off. We also analyze advanced high-dimensional optimization algorithms and suggest ways to tackle these challenges in analog circuit design. Additionally, we've launched an open-source benchmark for analog circuit optimization, available for research algorithm testing.

Acknowledgments

This research is supported by National Natural Science Foundation of China (NSFC) research projects 62141407, 62304052, 61974032, 61929102, 61674042, 62090025.

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